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# Exercise 3
# SQL CASE STATEMENTS

## 1. THE PRODUCT TABLE

Table: products

| Product_id | Product_name | Price |
|---|---|---|
| 1 | Laptop | 1200 |
| 2 | Phone | 800 |
| 3 | Keyboard | 45 |
| 4 | Monitor | 300 |
| 5 | Mouse | 25 |

Q1: Classify each product by price into three tiers
- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

ANSWER:

```
SELECT product_name, price,
    CASE
        WHEN price > 1000 THEN
        'Expensive'
        WHEN price BETWEEN 100 AND 1000
        THEN 'Mid-range'
        WHEN price < 100 THEN 'Budget'
    END AS price_category
FROM products;
```

TABLE: Product

| product_name | price | product_category |
|---|---|---|
| Laptop | 1200 | Expensive |
| Phone | 800 | Mid-range |
| Keyboard | 45 | Budget |
| Monitor | 300 | Budget |
| Mouse | 25 | Budget |

## 2. THE ORDERS TABLE

TABLE: Orders

| order_id | customer_name | amount |
|---|---|---|
| 1 | Alice | 150 |
| 2 | Bob | 560 |
| 3 | Charlie | 999.99 |
| 4 | Diana | 45.50 |
| 5 | Ethan | 1200 |

Q2: Label each order by its value
- 'High-value' for orders > 1000
- 'Medium value' for orders 500 - 999.99
- 'Low value' for orders < 500

ANSWER:

```
SELECT customer_name, amount,
    CASE
        WHEN order amount ≥ 1000 THEN 'High Value'
        WHEN order amount BETWEEN 500 AND 999.99
        THEN 'Medium Value'
        WHEN OR order amount < 500 THEN
        'LOW VALUE'
    END AS order_value_category
FROM orders;
```

TABLE: orders

| customer_name | amount | order-value |
|---|---|---|
| Alice | 150.00 | Low_Value |
| Bob | 560.00 | Medium_Value |
| Charlie | 999.99 | Medium_value |
| Diana | 45.50 | LOW_Value |
| Ethan | 1200.00 | High_Value |

## 3. THE EMPLOYEES TABLE

TABLE: employees

| emp_id | emp_name | department | salary |
|---|---|---|---|
| 1 | John | IT | 85000 |
| 2 | Sara | HR | 60000 |
| 3 | Mark | IT | 75000 |
| 4 | Lucy | Finance | 95000 |
| 5 | Tom | HR | 55000 |

Q3: Categorize each employee's position level using both department and salary
- If in 'IT' and salary > 80000 → 'Senior_IT'
- If in HR and salary > 55000 → 'Experienced HR'
- If in Otherwise → 'Staff'

ANSWER:

```
SELECT emp_name, department, salary,
    CASE
        WHEN department = 'IT' AND salary > 80000
        THEN 'Senior_IT'
        WHEN department = 'HR' AND salary > 55000
        THEN 'Experienced_HR'
        ELSE 'staff'
    END AS position_level
FROM employees;
```

Table: employees

| emp_name | department | salary | position_level |
|---|---|---|---|
| John | IT | 85000 | Senior_IT |
| Sara | HR | 60000 | Experienced_HR |
| Mark | IT | 75000 | Staff |
| Lucy | Finance | 95000 | Staff |
| Tom | HR | 55000 | Experienced_HR |

## 4. THE STUDENTS TABLE

TABLE: Students

| Student_Id | Student_name | Score |
|---|---|---|
| 1 | Anna | 92 |
| 2 | Ben | 76 |
| 3 | Cara | 59 |
| 4 | David | 83 |
| 5 | Elia | 68 |

Q4. Assign each student a letter grade based on their score

- >= 90 → 'A'
- 80-89 → 'B'
- 70-79 → 'C'
- 60-69 → 'D'
- <60 → 'F'

TABLE: students

| student_name | score | grade |
|---|---|---|
| Anna | 92 | |
| Ben | 76 | |
| cara | 59 | |
| David | 83 | |
| Elia | 58 | |



ANSWER:

```
SELECT student_name, score,
    CASE
        WHEN score >= A THEN 'A'
        WHEN score BETWEEN 80 AND 89 THEN 'B'
        WHEN score BETWEEN 70-79 THEN 'C'
        WHEN score BETWEEN 60-69 THEN 'D'
        WHEN score <60 THEN 'F'
    END AS grade
FROM students;
```

# 5. THE DELIVERIES TABLE

TABLE: deliveries

| delivery_id | delivery_time_minute |
|---|---|
| 1 | 45 |
| 2 | 80 |
| 3 | 30 |
| 4 | 65 |
| 5 | 100 |

Q5: Label delivery performance based on the time taken:
- ≤ 30 mins → 'fast'
- 31-60 mins → 'on time'
- > 60 mins → 'Late'

ANSWER:
```
SELECT *
    CASE
        WHEN minutes <= 30 THEN 'FAST'
        WHEN minutes BETWEEN 31 AND 60
        THEN 'ON TIME'
        WHEN minutes > 60 THEN 'LATE'
    END AS performance
FROM deliveries;
```

| delivery_id | delivery_time_minutes | Performance |
|---|---|---|
| 1 | 45 | ON TIME |
| 2 | 80 | LATE |
| 3 | 30 | Fast |
| 4 | 65 | LATE |
| 5 | 100 | LATE |

## 06. THE TICKETS TABLE

TABLE: tickets

| ticket_id | issue_type | Priority |
|---|---|---|
| 1 | Login issue | 1 |
| 2 | Server down | 3 |
| 3 | Slow system | 2 |
| 4 | Email error | 2 |
| 5 | Password reset | 1 |

Q6. Convert the numeric priority into a readable label.

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

ANSWER:
```
SELECT issue_type, Priority,
    CASE
        WHEN priority = 3 THEN 'High'
        WHEN priority = 2 THEN 'Medium'
        WHEN priority = 1 THEN 'Low'
    END AS priority_level
FROM tickets;
```

| ticket_id | issue_type | Priority | priority_label |
|---|---|---|---|
| 1 | Login issue | 1 | Low |
| 2 | Server down | 3 | High |
| 3 | Slow system | 2 | Medium |
| 4 | Email error | 2 | Medium |
| 5 | Password | 1 | Low |

## 7. THE ATTENDANCE TABLE

TABLE: attendance

| student_id | days_present | total_days |
|---|---|---|
| 1 | 45 | 50 |
| 2 | 30 | 50 |
| 3 | 48 | 50 |
| 4 | 25 | 50 |
| 5 | 50 | 50 |

Q7: Calculate attendance percentage and classify the result.

- * ≥ 90% → 'Excellent'
- * 75–89% → 'Good'
- * < 75% → 'Needs Improvent'

ANSWER:

```
SELECT
    student_id,
    (days_present * 100.0 / total_days) AS 'Attendance_perc',
    CASE
        WHEN (days_present * 100.0 / total_days) ≥ 90%
            THEN 'EXCELLENT'
        WHEN (days_present * 100.0 / total_days) BETWEEN
        75% AND 80% THEN 'Good'
        WHEN (days_absent * 100.0 / total_days) < 75%
            THEN 'Needs improvement'
    END AS attendance_status
FROM attendance
WHERE total_days > 0;
```

TABLE: attendance

| student_id | attendance_percentage | attendance_status |
|---|---|---|
| 1 | 90% | Excellent |
| 2 | 60% | Needs Improvement |
| 3 | 96% | Excellent |
| 4 | 50% | Needs Improvement |
| 5 | 100% | Excellent |

## 8. THE PRODUCTS_INVENTORY TABLE:

TABLE: products_inventory

| product_id | stock_qty |
|---|---|
| 1 | 5 |
| 2 | 0 |
| 3 | 25 |
| 4 | 10 |
| 5 | 3 |

Q8: Label the stock status of each product.
- 0 → 'out of stock'
- 1-5 → 'Low stock'
- >5 → 'In stock'

ANSWER:
```
SELECT *
    CASE
        WHEN stock quantity = 0 THEN 'out of stock'
        WHEN stock quantity BETWEEN 1 AND 5
          THEN 'Low stock'
        WHEN stock quantity >5 THEN 'IN Stock'
    END AS stock_status
FROM products_inventory;
```

| product_id | stock_qty | stock_status |
|---|---|---|
| 1 | 5 | IN stock |
| 2 | 0 | Out of stock |
| 3 | 25 | IN stock |
| 4 | 10 | IN stock |
| 5 | 3 | Low stock |

## 9. THE CLASS TABLE

TABLE: Classes

| class_id | Subject | enrolled_students |
|---|---|---|
| 1 | Math | 30 |
| 2 | English | 25 |
| 3 | Science | 15 |
| 4 | Art | 5 |
| 5 | History | 20 |

Q9: Classify each class by the number of enrolled students.

* ≥ 25 → 'Large'
* 10 - 24 → 'Medium'
* < 10 → 'Small'

ANSWER:

```
SELECT subject, enrolled_students,
    CASE
        WHEN enrolled students ≥ 25 THEN 'Large'
        WHEN enrolled students BETWEEN 10 AND 24
            THEN 'Medium'
        WHEN enrolled_students < 10 THEN 'SMALL'
    END AS class_size_category
FROM classes;
```

TABLE: classes

| Subject | enrolled_students | class_size_category |
|---|---|---|
| Math | 30 | Large |
| English | 25 | Large |
| Science | 15 | Medium |
| Art | 5 | Small |
| History | 20 | Medium. |

## 10. THE PAYMENTS TABLE.

TABLE: payments

| payment_id | amount | payment_method |
|---|---|---|
| 1 | 50.00 | Card |
| 2 | 200.00 | cash |
| 3 | 150.00 | Card |
| 4 | 75.00 | PayPal |
| 5 | 300.00 | cash |

Q10. Apply a discount flag based on the payment method and amount:

* If payment_method = 'cash' and amount ≥ 200 → 'eligible for discount'.
* Otherwise → 'Not eligible'

ANSWER:

```
SELECT payment_id, payment_method, amount,
    CASE
        WHEN payment_method = 'cash' AND amount ≥ 200
            THEN 'Eligible for discount'
        ELSE 'Not eligible'
    END AS discount_eligibility
FROM payments;
```

TABLE: payments

| payment_id | payment_method | amount | discount_eligibility |
|---|---|---|---|
| 1 | Card | 50.00 | Not eligible |
| 2 | cash | 200.00 | Eligible for discount |
| 3 | Card | 150.00 | Not eligible |
| 4 | PayPal | 75.00 | Not eligible. |
| 5 | cash | 300.00 | Eligible for discount |